392. Is Subsequence:

Given two strings `s` and `t`, return `true` *if* `s` *is a* ***subsequence*** *of* `t`, *or* `false` *otherwise*.

A **subsequence** of a string is a new string that is formed from the original string by deleting some (can be none) of the characters without disturbing the relative positions of the remaining characters. (i.e., `"ace"` is a subsequence of `"abcde"` while `"aec"` is not).

Example 1:

```
Input: s = "abc", t = "ahbgdc"
Output: true
```

Example 2:

```
Input: s = "axc", t = "ahbgdc"
Output: false
```

Constraints:

- `0 <= s.length <= 100`
- `0 <= t.length <= 104`
- `s` and `t` consist only of lowercase English letters.

Solution: Time = O(n) and Space = O(m+n)

```python
class Solution:
    def isSubsequence(self, s: str, t: str) -> bool:
        s = list(s)
        t = list(t)
        #pointers
        l = 0
        r = 0
        ans = True
```

```
        while l < len(s):
            if r > len(t)-1:
                return False
            elif s[l] == t[r]:
                l += 1
                r += 1
                if r > len(t):
                    ans = False
                ans = True
            elif s[l] != t[r]:
                r += 1
        return ans
```

Optimal solution: Time = O(n) and Space = O(1):

```
class Solution:
    def isSubsequence(self, s: str, t: str) -> bool:

        l = 0
        r = 0

        while l < len(s) and r < len(t):

            if s[l] == t[r]:
                l += 1

            r += 1

        return l == len(s)
```